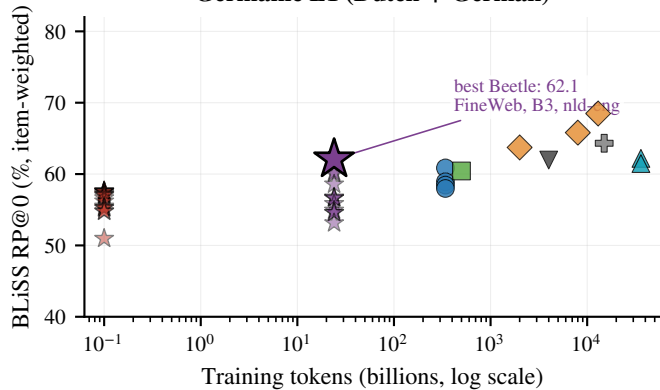
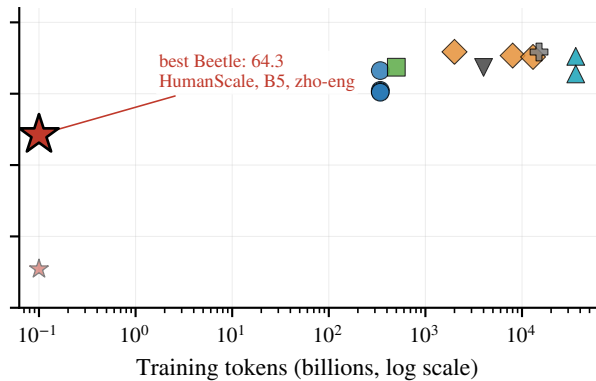


Beetle approaches frontier-LLM BLiSS L2 alignment with ≤ 0.2 % of the training data

Germanic L1 (Dutch + German)



Chinese L1



- ★ Beetle-HumanScale (150 M, 0.1 B tok)
- ★ Beetle-FineWeb (150 M, 24 B tok)

- BLOOM
- XGLM

- ◆ Gemma-2
- ⊕ Llama-3.1

- ▲ Qwen-3
- ▼ EuroLLM